

Question ID: 3d7d7534

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$(d - 30)(d + 30) - 7 = -7$

What is a solution to the given equation?

Correct Answer: 30, -30

Rationale

The correct answer is either **−30** or **30**. Adding **7** to each side of the given equation yields $(d - 30)(d + 30) = 0$. Since a product of two factors is equal to **0** if and only if at least one of the factors is **0**, either $d - 30 = 0$ or $d + 30 = 0$. Adding **30** to each side of the equation $d - 30 = 0$ yields $d = 30$. Subtracting **30** from each side of the equation $d + 30 = 0$ yields $d = -30$. Therefore, the solutions to the given equation are **−30** and **30**. Note that -30 and 30 are examples of ways to enter a correct answer.